FJRW-ThreeBlock-F: Problem Statement and Reconstruction Boundary Tables for the Three-Block Thom-Sebastiani Sum F = W + W + W

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/FJRW-ThreeBlock-F
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Abstract

Standalone dataset and formal problem specification (‘docs/PROBLEM.md‘, ‘docs/RECONSTRUCTION_BOU ‘docs/WHAT_THIS_IS_NOT.md‘, ‘tables/block_comparison.csv‘, ‘tables/not_closed_by_concavity.csv‘) for the Fan-Jarvis-Ruan-Witten (FJRW) cohomological field theory of the three-block Thom-Sebastiani sextic polynomial F = W (+) W (+) W with chain atom W = u^5 v + v^6.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Defines the three-block FJRW problem and explicitly tabulates the correlators that lie beyond pure concavity axioms; does not claim an unproved closed-form evaluation of open mixed correlators.

- **Repository:** `https://github.com/BenFrohman/FJRW-ThreeBlock-F`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652984`
- **Earliest Git Commit SHA:** `515ac1028fa0c5a4f678e7b47221c7d7244fbbd9` (2026-09-25 03:31:31 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman (@BenFrohman)
Copyright (c) 2026 Benjamin Stanley Frohman. Apache-2.0. Public.
This repo states one problem. It does not solve it.
Atom lab: [BenFrohman/SingularityLab](https://github.com/BenFrohman/SingularityLab)
W = u^5 v + v^6 = X^5 Y + Y^6
(W) = 25, |Aut(W)| = 30, ĉ = 4/3
Guéré's chain formula applies to this W. Fan–Shen labels (p,q)=(5,6), gcd(4,6)=2.
F = W(x0,x3) + W(x1,x4) + W(x2,x5) P^5
Jac(F) Jac(W)^{ 3} dim 15625
Aut(F) Aut(W)^3 order 27000
ĉ(F) = 4

State spaces tensor. Correlators do not.
See PROBLEM.md.
No Guéré table is claimed. No Hodge miss. Not Term B.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/FJRW-ThreeBlock-F` is audited via `titan-audit` (`audit/TITAN_AUDIT_CE`
and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`)
signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	329 B	4ae3fd4ee9a1e1b8...
PROBLEM.md	2945 B	daa906cefad1494d...
README.md	778 B	49b525fe8a776cba...
RECONSTRUCTION.md	2376 B	d1a4cee1c06f4729...
RECONSTRUCTION_THEOREMS.md	3260 B	7709d34699902029...
RING_RECONSTRUCTION.md	2529 B	08ca8ec2d8d7f467...
TWO_GROUPS.md	1996 B	5d70d8d27bc0d784...
docs/ALL_CLOSED_CLAIMS.md	2411 B	5d03032985b22cc2...
docs/CLOSED_CLAIMS.md	2343 B	7858201a5a9f63d9...
docs/HILBERT_VS_HODGE.md	2106 B	54e680517c002638...
docs/J_SECTORS.md	3345 B	605d7f4fbb4b3df5...
docs/STATE_SPACE_VS_CORRELATORS.md	662 B	2b30e16509529b88...
docs/WHAT_FACTORS.csv	502 B	78d93e8e0f84eebd...
docs/tables/claim_status.csv	582 B	92a6615795e054a8...
docs/tables/closed_claims.csv	529 B	5621a38275619a5d...
docs/tables/j_sectors.csv	199 B	85cc759c3d05675d...
docs/tables/three_points_W_full.csv	652 B	ad3c45d336490b67...
docs/three_points_W.csv	602 B	58a50bf7a2587329...
docs/three_points_W_seed.csv	128 B	af704fb95d9a52ec...

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